COMMUNICATION IN BLACK SPOT REGIONS USING LoRa MESH WITH EMSRA ROUTING

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ABSTRACT: Modern communication systems fail in black spot regions such as forests, tunnels, highways, border areas, and disaster zones due to lack of network infrastructure. This results in delayed emergency response and increased risk to human life. The proposed system introduces a LoRa mesh-based emergency communication framework integrated with an Emergency Mesh Signal Routing Algorithm (EMSRA) to enable reliable multi-hop communication. The system is developed using ESP32 controllers, SX1278 LoRa modules, GPS tracking, and crash detection sensors. It supports intelligent SOS triggering, real-time location sharing, and offline rescue mapping without relying on cellular networks. The inclusion of solar-powered Li-ion nodes ensures long-term autonomous deployment. The system achieves improved packet delivery ratio (up to 94%), enhanced communication reliability, and reduced emergency response time, making it highly suitable for infrastructure-less environments such as disaster zones, forests, and border areas.

Keywords: LoRa Mesh, LPWAN, Emergency Communication, EMSRA, GPS Tracking, IoT, WSN.

I. INTRODUCTION

Communication plays a vital role in emergency response, disaster management, and public safety. However, several regions such as forests, tunnels, hilly terrains, mines, highways, and border areas suffer from network black spots, where communication infrastructure is either weak or unavailable.

Traditional communication systems such as GSM, Wi-Fi, and Bluetooth rely heavily on infrastructure like towers and access points, making them ineffective in remote environments. These limitations lead to delayed rescue operations, increased risks, and reduced efficiency of emergency services. LoRa (Long Range) technology, a part of Low Power Wide Area Networks (LPWAN), offers long-distance communication with minimal power consumption, making it ideal for remote applications. However, existing LoRa systems are

mostly limited to single-hop communication, which reduces scalability and reliability.
To overcome these challenges, this paper proposes a LoRa mesh-based communication system integrated with EMSRA routing. The system enables multi-hop communication, GPS-based tracking, and intelligent emergency alert transmission, ensuring reliable communication in black spot regions.

II. LITERATURE REVIEW

The use of LoRa and IoT technologies for communication in remote areas has gained significant attention. Various systems have been developed to improve long-range communication; however, several limitations still exist. In [1], Kanaka Raja et al. proposed a LoRa-based communication system using GSM forwarding, which enables message transmission from remote areas. However, the system depends on GSM connectivity and lacks scalability.
In [2], Padmaja et al. developed a LoRa communication system for restricted areas, but it lacks GPS integration and automation features.
In [3], Durand and Booysen evaluated a mesh topology LoRa network and demonstrated that packet delivery ratio improved from 40.2% to 73.78%, proving the effectiveness of multi-hop communication. However, the study was simulation-based.
In [4], Sundaram et al. presented a survey on LPWAN technologies but did not address real-world deployment challenges.
In [5], Escobar et al. proposed a multi-hop protocol but did not evaluate energy efficiency for large-scale deployments.
These studies highlight the need for a reliable, scalable, and intelligent communication system specifically designed for black spot emergency scenarios.
The paper aims to achieve the following objectives:

- Develop a LoRa mesh-based communication system for black spot regions.
- Enable multi-hop emergency SOS transmission.
- Integrate GPS tracking and crash detection for real-time alerts.
- Implement solar-powered autonomous nodes.
- Improve packet delivery ratio and reduce communication delay.

III. METHODOLOGY

The proposed system is designed as a decentralized LoRa mesh-based emergency communication framework to enable reliable communication in network black spot regions. The methodology integrates hardware components, communication protocols, and an intelligent routing algorithm (EMSRA) to ensure efficient and continuous operation.

A. System Overview

The system operates by detecting emergency situations either manually through an SOS trigger or automatically using sensor-based detection. Once an emergency is identified, the system collects GPS location and sensor data, encodes it into a data packet, and transmits it through a LoRa mesh network using multi-hop communication.
Each node in the network acts as a transmitter, receiver, and repeater, enabling the system to extend communication range without relying on infrastructure such as cellular networks or internet connectivity.

B. Node Deployment

LoRa nodes are strategically deployed in black spot regions such as forests, highways, tunnels, and hilly terrains to ensure maximum coverage. Nodes are placed at elevated positions (trees, poles, towers) to reduce signal obstruction. Deployment density varies based on terrain: dense deployment is used in forests and mines, while sparse deployment is preferred for highways and open areas. Each node forms part of a self-configuring mesh network, ensuring redundancy and reliability even if some nodes fail.

C. Emergency Detection Mechanism

The system supports both manual and automatic emergency detection:

1) Manual Detection: The user presses the SOS button, generating an immediate alert.
2) Automatic Detection: An accelerometer continuously monitors motion and detects sudden impacts, crash events, and abnormal vibrations. If predefined thresholds are exceeded, the system automatically triggers an emergency alert without user intervention.

D. Data Acquisition

Once an emergency is detected, the ESP32 microcontroller collects and processes the following data: GPS coordinates (latitude and longitude), timestamp of the event, sensor readings (impact level), and emergency type.

E. LoRa Mesh Communication

The communication system is based on multi-hop LoRa mesh networking. The source node generates an emergency packet, which is broadcast to neighboring nodes. Intermediate nodes receive and forward the packet until it reaches the rescue node. Key features include long-range communication (10–20 km), low power consumption, and infrastructure-independent operation. This approach significantly improves communication reliability compared to single-hop systems.

F. EMSRA Routing Algorithm

The Emergency Mesh Signal Routing Algorithm (EMSRA) is designed to ensure reliable and efficient packet transmission in the mesh network.

Input: Sensor data, GPS coordinates, SOS signal. Output: Emergency message delivered to rescue station. The algorithm executes the following steps:

1) Initialize all LoRa nodes in the network.
2) Detect the emergency event.
3) Generate a high-priority emergency packet.
4) Broadcast the packet to neighboring nodes.
5) For each receiving node, check whether the packet has already been processed; if yes, discard it; otherwise, forward it.
6) Select the next hop based on signal strength, distance, and node availability.
7) Continue transmission until the destination is reached.
8) Confirm delivery and terminate propagation.

EMSRA features include priority-based routing, duplicate packet elimination, efficient path selection, and reduced network congestion.

G. Software Requirements

The system is implemented using Arduino IDE for ESP32 programming. The software continuously monitors sensor inputs, processes GPS data, and transmits information via LoRa. Data handling includes encoding, transmission, and error checking.

H. Hardware Requirements

The hardware system consists of ESP32 microcontrollers, LoRa (SX1278) modules, GPS receivers, accelerometer sensors, and solar-powered Li-ion power units. Nodes are deployed strategically in black spot areas to ensure maximum coverage. Solar panels provide continuous charging, creating a self-sustaining power architecture that keeps the network functional in remote, infrastructure-poor environments.

IV. RESULTS AND DISCUSSION

The proposed system improves communication reliability and coverage in black spot regions. Compared to single-hop systems, the mesh network increases the packet delivery ratio (up to 94%), reduces transmission failure, extends communication range, and enables real-time emergency alerts. The system performs effectively in remote environments such as forests, highways, and disaster zones.

Key Advantages

- Operates without internet or GSM infrastructure.
- Long-range communication (10–20 km).
- Low power consumption.
- Scalable mesh architecture.
- Real-time GPS tracking.
- Autonomous operation using solar power.

Target Applications

- Disaster management systems.
- Forest monitoring and rescue.
- Border security communication.
- Highway accident alert systems.
- Military operations.
- Remote area communication.

V. CONCLUSION

The proposed LoRa mesh-based communication system provides a reliable and scalable solution for black spot regions. By integrating EMSRA routing, GPS tracking, and solar-powered nodes, the system ensures efficient emergency communication without relying on existing infrastructure. The system significantly improves response time and enhances safety in remote and disaster-prone environments.

FUTURE SCOPE

- AI-based intelligent routing optimization.
- Drone-assisted communication nodes.
- Integration with satellite communication.
- Mobile application with real-time tracking dashboard.

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